Busundo - Product & Development Plan

Version: 1.0

Domain: busundo.com

Tagline: "Bus Undo? Open Busundo."

Executive Summary

Busundo is a community-powered bus travel platform designed for students, office workers, and daily commuters.

Unlike traditional bus apps that only provide schedules and routes, Busundo focuses on helping users coordinate travel with friends, classmates, and other commuters.

The platform will initially launch in Kerala and later expand to other regions.

Core Mission:

Help users answer:

- Is there a bus available?
- Which route should I take?
- Who else is travelling?
- Can I travel together with friends?
- Is the bus delayed or crowded?

Problem Statement

Current Problems:

1. Bus schedules are difficult to find.
2. Route information is fragmented.
3. Students cannot coordinate travel.
4. New commuters don't know which bus to take.
5. Real-time information is missing.

Busundo solves these problems through crowd-sourced travel coordination.

Unique Selling Proposition

Busundo is not a bus tracking app.

Busundo is a community travel platform.

Traditional Apps:

- Routes
- Timings

Busundo:

- Routes
- Travel Groups
- Friend Coordination
- Community Updates
- Crowd Reports

Target Audience

Primary Users

- College Students
- School Students
- Office Employees

Secondary Users

- Tourists
- Daily Commuters
- New Residents

Launch Audience

- Kerala College Students

Technology Stack

Frontend:

- Next.js 15

- TypeScript
- Tailwind CSS
- Shadcn UI

Backend:

- Next.js API Routes
- Server Actions

Database:

- MongoDB Atlas Cluster
- Mongoose

Authentication:

- NextAuth.js
- Google Login

State Management:

- Zustand

Validation:

- Zod

Realtime:

- Socket.IO

Maps:

- OpenStreetMap
- Leaflet

Hosting:

- Vercel

Media Storage:

- Cloudinary

Analytics:

- Vercel Analytics

Architecture

Client

Next.js Frontend

↓

Server Actions / API Routes

↓

MongoDB Atlas

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Socket.IO

↓

Real-Time Updates

MVP Features

Authentication

Features:

- Google Login
- User Profile
- Session Management

User Fields:

- Name
- Email
- Avatar

Route Search

User Inputs:

- Source
- Destination

System Returns:

- Matching Routes
- Stops
- Estimated Duration

Travel Groups

Users can create travel groups.

Example:

Aluva → Kakkanad 8:30 AM

Group Details:

- Route
- Time
- Members
- Status

Join Bus

User clicks:

Join This Bus

System:

- Adds user to group
- Updates member count

Travel Status

Available Status:

- Waiting
- Reached Stop
- Boarded Bus
- Delayed
- Journey Completed

Notifications

Notify Users When:

- Group Created
- New Member Joined
- Status Updated

Share Route

Generate Public Link

Example:

https://busundo.com/group/abc123

Share Via:

- WhatsApp
- Facebook
- Instagram
- Telegram

Phase 2 Features

Group Chat

Temporary travel chat.

Auto expires after journey.

Friend Coordination

Example:

3 students from MEC travelling.

Join same bus?

Bus Delay Reports

Community generated reports.

Examples:

- Delayed
- Cancelled
- Full

College Communities

Dedicated communities.

Examples:

- MEC
- CUSAT
- CET
- Rajagiri

Favorite Routes

Users save routes.

One click access.

Phase 3 Features

Live Crowd Tracking

Users optionally share location.

Crowd estimates bus movement.

AI Recommendations

Suggest:

- Fastest Route
- Less Crowded Route
- Cheapest Route

Smart Travel Suggestions

Example:

Your friend Arun is travelling on this route.

Would you like to join?

Database Collections

users

- _id
- name
- email
- image
- college
- favoriteRoutes
- createdAt
- updatedAt

---

routes

- _id
- source
- destination
- stops
- duration
- createdAt

---

groups

- _id
- routeId
- creatorId
- date
- time
- memberCount
- status
- createdAt

---

groupMembers

- _id
- groupId
- userId
- travelStatus
- joinedAt

---

notifications

- _id
- userId
- title
- message
- read
- createdAt

---

reports

- _id
- groupId

- userId
- type
- description
- createdAt

Folder Structure

src/

app/ (auth)/ dashboard/ routes/ groups/ profile/ api/

components/ ui/ routes/ groups/ shared/

actions/

lib/ mongodb/ auth/ socket/ utils/

models/

services/

hooks/

store/

types/

constants/

API Endpoints

Authentication

POST /api/auth/login

POST /api/auth/logout

GET /api/auth/session

Routes

GET /api/routes

GET /api/routes/search

GET /api/routes/[id]

Groups

POST /api/groups

GET /api/groups

GET /api/groups/[id]

POST /api/groups/join

DELETE /api/groups/leave

Status

POST /api/status/update

GET /api/status/group/[id]

Notifications

GET /api/notifications

PATCH /api/notifications/read

Pages

Landing Page

Features:

- Hero Section
- Search CTA
- App Benefits
- Testimonials

- Download CTA

Dashboard

Features:

- Search Route
- Recent Routes
- Active Groups

Group Page

Features:

- Member List
- Status Updates
- Share Button

Profile Page

Features:

- User Info
- Favorite Routes
- Joined Groups

Monetization

Stage 1

Free

Focus on user acquisition.

Stage 2

Sponsored listings.

Local businesses.

Stage 3

Premium Subscription.

Features:

- Ad Free
- Priority Alerts
- Enhanced Route Insights

Success Metrics

Month 1

100 Users

Month 3

1000 Users

Month 6

10000 Users

Development Roadmap

Week 1

- Setup Next.js
- Setup MongoDB
- Setup NextAuth
- Design System

Week 2

- Authentication
- User Profiles
- Landing Page

Week 3

- Route Search
- Route Details

Week 4

- Travel Groups
- Join Bus

Week 5

- Notifications
- Sharing Features

Week 6

- Testing
- Bug Fixes
- Production Deployment

Important Principle

Do not attempt live bus tracking initially.

Focus on:

1. Route Discovery
2. Travel Groups
3. Sharing
4. Community Building

The first goal is to validate user behavior, not to build a perfect transport system.